AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

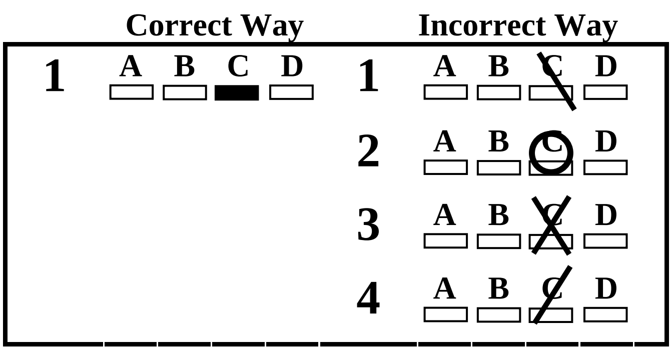
CLASS XI EXAMINATION 2009

Physics Paper I

Time allowed: 40 minutes Marks 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

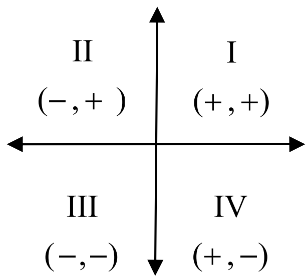
1. A Pascal, the S.I. Unit of pressure, is

A. $kg.m^{-1}.S^{-2}$.
B. $kg.\ m^{-1}.S^{-1}$.
C. $kg\ m^{-1}.S$.
D. kg m.S.

2. Which of the following shows the dimensions of force, velocity and acceleration respectively?

A. $MLT^{-2}, LT^{-1}, LT^{-2}$
B. $MLT^{-2}, LT^{-1}, LT^{2}$
C. $ML^{2}T, LT^{-1}, LT^{-2}$
D. MLT^{-2}, LT^{2}, LT

3. If R_x and $R_{y,}$ components of a vector, are both negative then the resultant lies in the

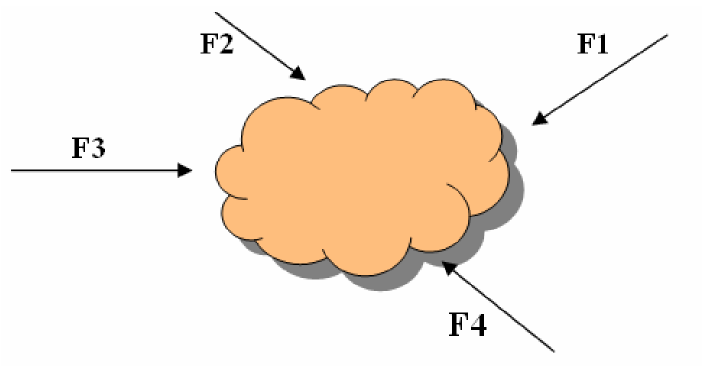


A. Third quadrant and its direction is $\theta = 180^0 - \theta$.
B. Third quadrant and its direction is $\theta = 180^0 + \theta$.
C. Second quadrant and its direction is $\theta = 180^0 - \theta$.
D. Second quadrant and its direction is $\theta = 180^0 + \theta$.

4. The scalar product of $\hat{i}$ and $\hat{j}$ is

A. zero
B. one
C. $-\hat{k}$
D. $\hat{k}$

5. A body is in equilibrium under the action of four forces shown below. State the shape of their head to tail diagram.



A. Square
B. Rectangle
C. Quadrilateral
D. Parallelogram

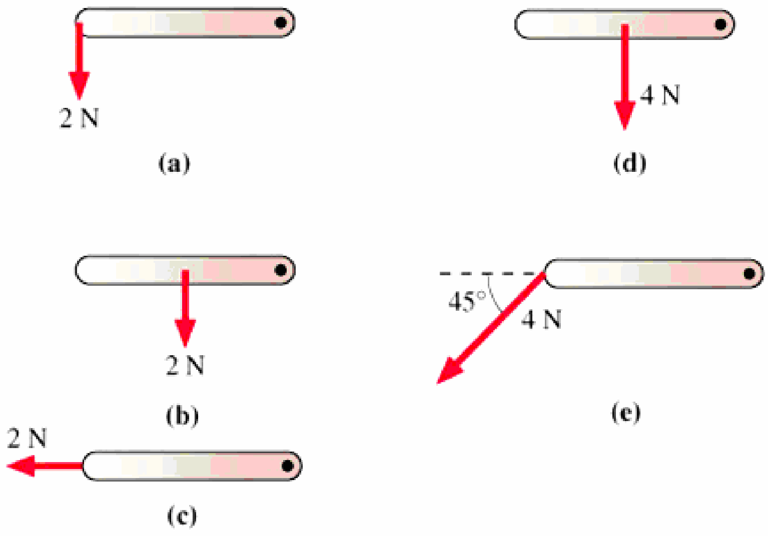
6. When a system of two bodies undergoes an elastic collision, then

A. P.E and momentum remain conserved.
B. K.E and momentum remain conserved.
C. K.E remains conserved and momentum changes.
D. Momentum remains conserved and K.E changes.

7. The horizontal range of a projectile is given by $R = \frac{V_i^2}{g}\text{Sin}2\theta$. It is maximum at an angle of

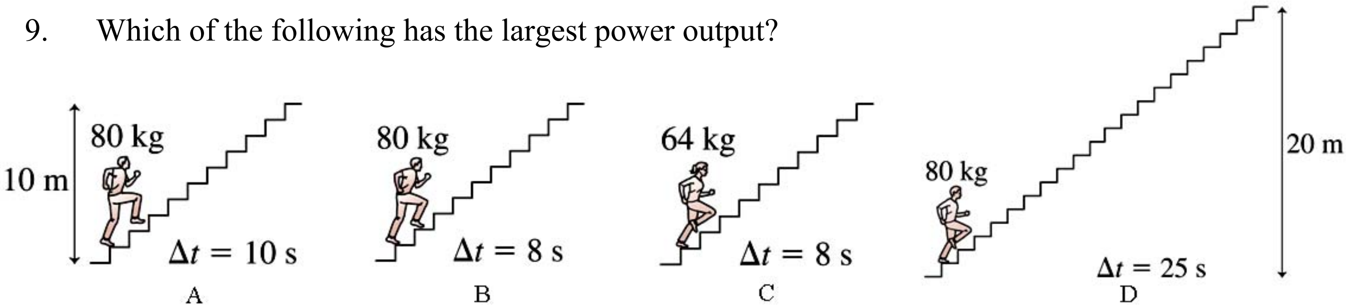
A. 0°
B. 30°
C. 45°
D. 60°

8. Which of the following represents decreasing order of torque in five rods of the same length pivoted at the dot?



A. $\check{T}_e > \check{T}_a = \check{T}_d > \check{T}_b > \check{T}_c$
B. $\check{T}_d = \check{T}_e > \check{T}_a = \check{T}_b = \check{T}_c$
C. $\check{T}_d > \check{T}_e > \check{T}_a = \check{T}_b > \check{T}_c$
D. $\check{T}_d = \check{T}_e > \check{T}_d = \check{T}_b > \check{T}_c$

9. Which of the following has the largest power output?



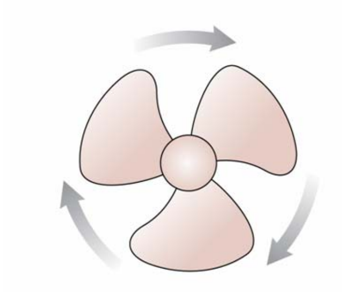
10. If the moon's radius is 1600 km and 'g' on its surface is 1.6 ms^{-2}, then the escape velocity on the moon is

A. 1882 ms^{-1}
B. 2000 ms^{-1}
C. 2263 ms^{-1}
D. 2600 ms^{-1}

11. In an isolated system, the conservation of mechanical energy states that the sum of

A. kinetic and potential energies remains constant.
B. kinetic and wave energies remains constant.
C. kinetic and tidal energies remains constant.
D. kinetic and heat energies remains constant.

12. The fan blade is slowing down. What are the signs of α and ώ?

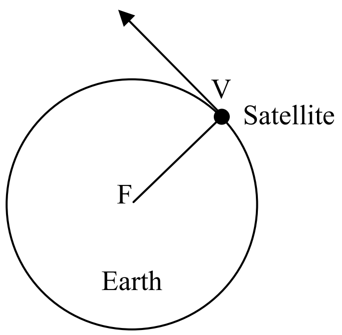


A. α is positive and ώ is positive
B. α is negative and ώ is positive
C. α is positive and ώ is negative
D. α is negative and ώ is negative

13. When a car of mass 500kg is moving around a turning path of radius 50m with 10m/s velocity, then the force of friction is

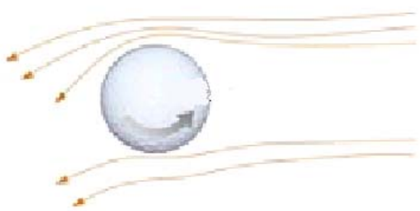
A. 125000 N
B. 25000 N
C. 1000 N
D. 560 N

14. The velocity which keeps a satellite in orbit is known as



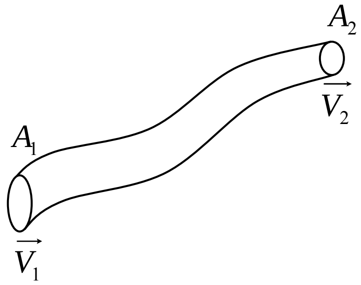
A. artificial velocity.
B. angular velocity.
C. escape velocity.
D. orbital velocity.

15. If a cricket ball is rotating on its axis as shown in the figure below, what is the direction of its movement?



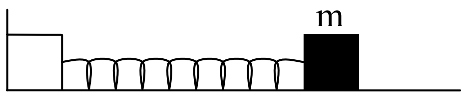
A. Downward
B. Rightward
C. Leftward
D. Upward

16. The equation $A_1V_1 = A_2V_2$ is called



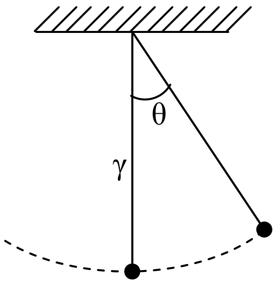
A. Equation of continuity.
B. Bernoulli's equation.
C. Venturi relation.
D. Stoke's law.

17. In a simple harmonic motion, the acceleration of the body at any instant is directly proportional to



A. amplitude.
B. displacement.
C. applied force.
D. mean position.

18. The time period of a simple pendulum is independent of



A. acceleration due to gravity.
B. length of pendulum.
C. mass suspended.
D. the angle θ.

19. While crossing a suspension bridge, soldiers are ordered to break their marching step to avoid

A. Doppler's Effect.
B. interference.
C. resonance.
D. beats.

20. The process in which energy is dissipated from an oscillating system is known as

A. phase.
B. damping.
C. resonance.
D. simple harmonic motion.

21. The observed change in the frequency due to the relative motion between the source and the observer is called

A. resonance.
B. Compton effect.
C. Doppler’s effect.
D. ultrasonic effect.

22. When a source of sound moves towards a stationary listener, there occurs

A. an increase in wavelength.
B. an increase in frequency.
C. a decrease in frequency.
D. no change in frequency.

23. Interference of light can be observed by using light of

A. different intensities.
B. the same intensity but from different sources.
C. the same wave length.
D. the same wave length but different frequencies.

24. In diffraction Grating’s formula ‘d’ is used for distance between

A. two slits.
B. slits and screen.
C. two dark fringes.
D. the centre and the nth fringe.

25. Polarization of light confirms the

A. wave nature of light.
B. particle nature of light.
C. transverse nature of light.
D. longitudinal nature of light.

26. The formula for pressure exerted by a gas on the walls of a container is

A. $\frac{1}{3} m \overline{V}^2$
B. $\frac{1}{3} m N_o \overline{V}^2$
C. $\frac{2}{3} m N_o \overline{V}$
D. $\frac{2}{3} m N_o \overline{V}^2$

27. Change in internal energy is due to the

    A. molecular P.E.
    B. molecular P.E & K.E.
    C. molecular rotational K.E.
    D. molecular translational K.E.

28. Molar specific heat at constant volume is the application of

    A. isobaric process.
    B. isochoric process.
    C. adiabatic process.
    D. isothermal process.

29. The equation for adiabatic process of 1st law of thermodynamics is

    A. $\Delta Q = \Delta U + \Delta w$
    B. $\Delta U = -\Delta w$
    C. $\Delta Q = \Delta w$
    D. $\Delta Q = \Delta U$

30. According to 2nd law of thermodynamics, heat can be converted into mechanical work if the system contains

    A. two heat reservoirs at different temperatures.
    B. engine and heat reservoir at zero temperature.
    C. engine and heat reservoir at the same temperature.
    D. engine and two heat reservoirs at different temperatures.

END OF PAPER